

# Educational Analytics Intelligence Report

## An End-to-End Analysis of Student Academic Performance, Attendance, and Learning Outcomes

**Prepared by:**  Person Kamal Ibrahim

**Institution:** University of the People

**Date:**  Date

### Executive Summary

Educational institutions generate substantial volumes of academic and administrative data every academic term. However, these data often remain underutilized, limiting their potential to support evidence-based educational decision-making. This project demonstrates how modern data analytics techniques can transform raw educational data into meaningful insights that support school administrators, teachers, and policymakers.

The project followed a complete end-to-end analytics workflow beginning with database design in MySQL, followed by cloud deployment to Microsoft Azure Database for MySQL, SQL querying and view creation, data cleaning and exploratory analysis using Python, and the development of an interactive Power BI dashboard connected to the cloud-hosted database. This architecture demonstrates how modern educational analytics solutions can integrate database management, cloud computing, programming, and business intelligence into a single decision-support platform. The analysis examined student attendance, academic performance, and progression across three academic terms. Findings indicate that while students maintained high attendance rates (over 92%), attendance alone was only moderately associated with academic achievement. Subject-specific analysis revealed that Religious and Moral Education (RME) consistently recorded the highest scores, while Mathematics produced the lowest. A gradual decline in performance was noted across successive terms, suggesting a need for mid-year instructional intervention.

# 1. Project Background

Educational institutions increasingly rely on data to evaluate student learning and improve instructional quality. Despite the availability of extensive data repositories—including examination scores and attendance logs—many schools in Ghana still treat this information as an administrative afterthought.

Educational analytics addresses this challenge by transforming raw institutional data into actionable insights through statistical analysis and interactive visualization. By combining database technologies (SQL), programming languages (Python), and business intelligence tools (Power BI), administrators can monitor student outcomes and identify high-risk learners more effectively. This project develops a comprehensive solution to analyze student performance across three academic terms, demonstrating a complete end-to-end data analytics workflow.

## 2. Project Objectives

The primary objective of this project was to develop a comprehensive educational analytics solution to support data-driven decision-making. Specifically, the project aimed to:

- Design and organize an educational database using MySQL.
- Clean and prepare educational datasets for analysis.
- Analyze student academic performance and attendance patterns across multiple terms.
- Compare subject performance to identify academic strengths and weaknesses.
- Examine the relationship between attendance and academic achievement.
- Develop an interactive Power BI dashboard for real-time monitoring.

## 3. Dataset Overview

The dataset represents student academic records collected across three academic terms, organized using a dimensional model consisting of fact and dimension tables.

| Dataset                 | Description                                         |
|-------------------------|-----------------------------------------------------|
| Student Dimension       | Student identification and demographic information. |
| Academic Term Dimension | Academic calendar containing three school terms.    |
| Attendance Records      | Longitudinal student attendance across each term.   |
| Student Performance     | Subject examination scores and academic averages.   |

## 4. Methodology

This project followed a structured data analytics workflow to transform raw educational records into meaningful insights for decision-making.

### 4.1 . Data Collection

Student academic and attendance data were collected from multiple assessment sources, including examination scores, class tests, group work, and attendance records. The dataset contained student information, subject scores, attendance records, and academic terms.

### 4.2. Data Warehouse Design

A dimensional data warehouse was designed using MySQL to support efficient analytical queries. The database was modeled using a star schema consisting of:

- **Fact Tables:** Fact\_ExamScore, Fact\_Test2, Fact\_GroupWork, Fact\_Attendance
- **Dimension Tables:** Dim\_Student, Dim\_Term

Views were created to simplify reporting by combining information from multiple tables. These included:

- vw\_student\_performance

- vw\_attendance\_summary
- vw\_subject\_scores

### 4.2.1 Cloud Database Deployment

To simulate a real-world enterprise analytics environment, the MySQL database was deployed to Microsoft Azure Database for MySQL. Hosting the database in Azure enabled centralized storage of educational records while allowing secure remote access from SQL clients, Python notebooks, and Microsoft Power BI. Rather than relying on a local database instance, the cloud deployment demonstrates how educational institutions can manage scalable databases that support multiple analytical tools simultaneously. The Azure-hosted database served as the primary data source throughout the project.

## 4.3. Data Cleaning and Transformation

Python (Pandas) was used to clean and prepare the data for analysis. The preparation process included:

- Handling missing values
- Standardizing column names
- Correcting data types
- Calculating overall student averages
- Computing attendance percentages based on the total instructional days in each academic term
- Creating performance bands and attendance categories
- Validating transformed data against SQL outputs

## 4.4 Data Preparation

The collected datasets were imported into a MySQL relational database and organized using a dimensional model consisting of fact and dimension tables. Student records, attendance information, assessment results, and academic term data were integrated through common identifiers such as Student\_ID and Term\_ID to ensure data consistency across the database.

To support efficient analysis, SQL views were created to consolidate data from multiple tables into reporting-ready datasets. These views simplified downstream analysis by providing a single source for student performance, attendance summaries, and

subject-level performance metrics, which were subsequently imported into Python and Power BI.

Database relationships and data integrity were validated throughout this stage to ensure that all records were correctly linked before analytical processing began.

## **4.5 Exploratory Data Analysis (EDA)**

Exploratory Data Analysis (EDA) was conducted using Python (Pandas, Matplotlib, and Seaborn) to examine the distribution of student scores, identify trends across academic terms, evaluate attendance patterns, and investigate factors associated with academic performance. The analysis provides a statistical foundation for the insights presented in the Power BI dashboard.

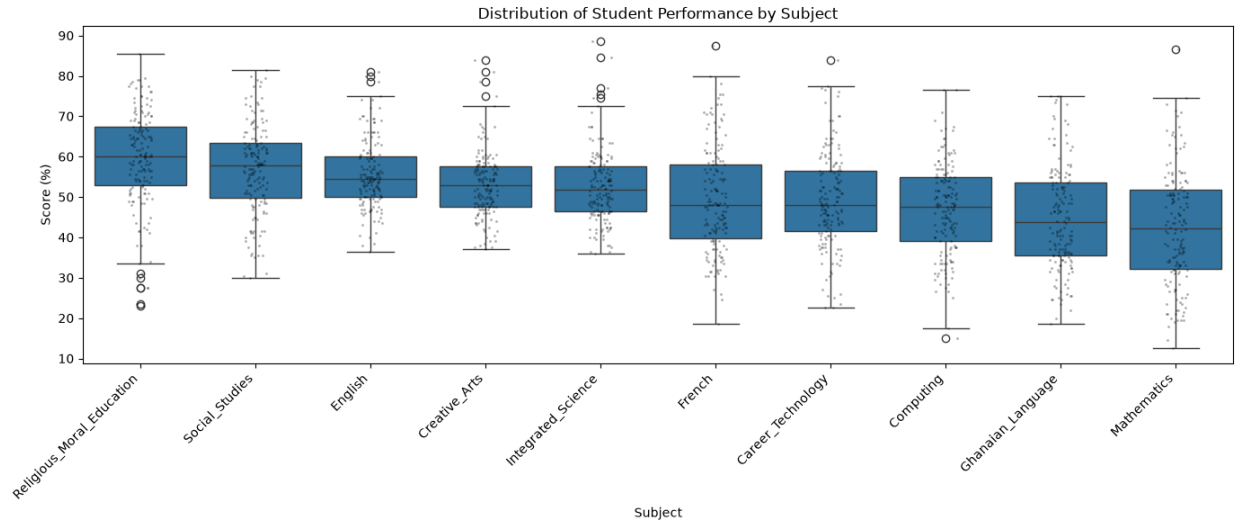
- Distribution of subject performance
- Performance trends across academic terms
- Attendance patterns
- Correlation between attendance and academic performance
- Subject correlation with overall academic achievement
- Identification of top- and low-performing students and subjects

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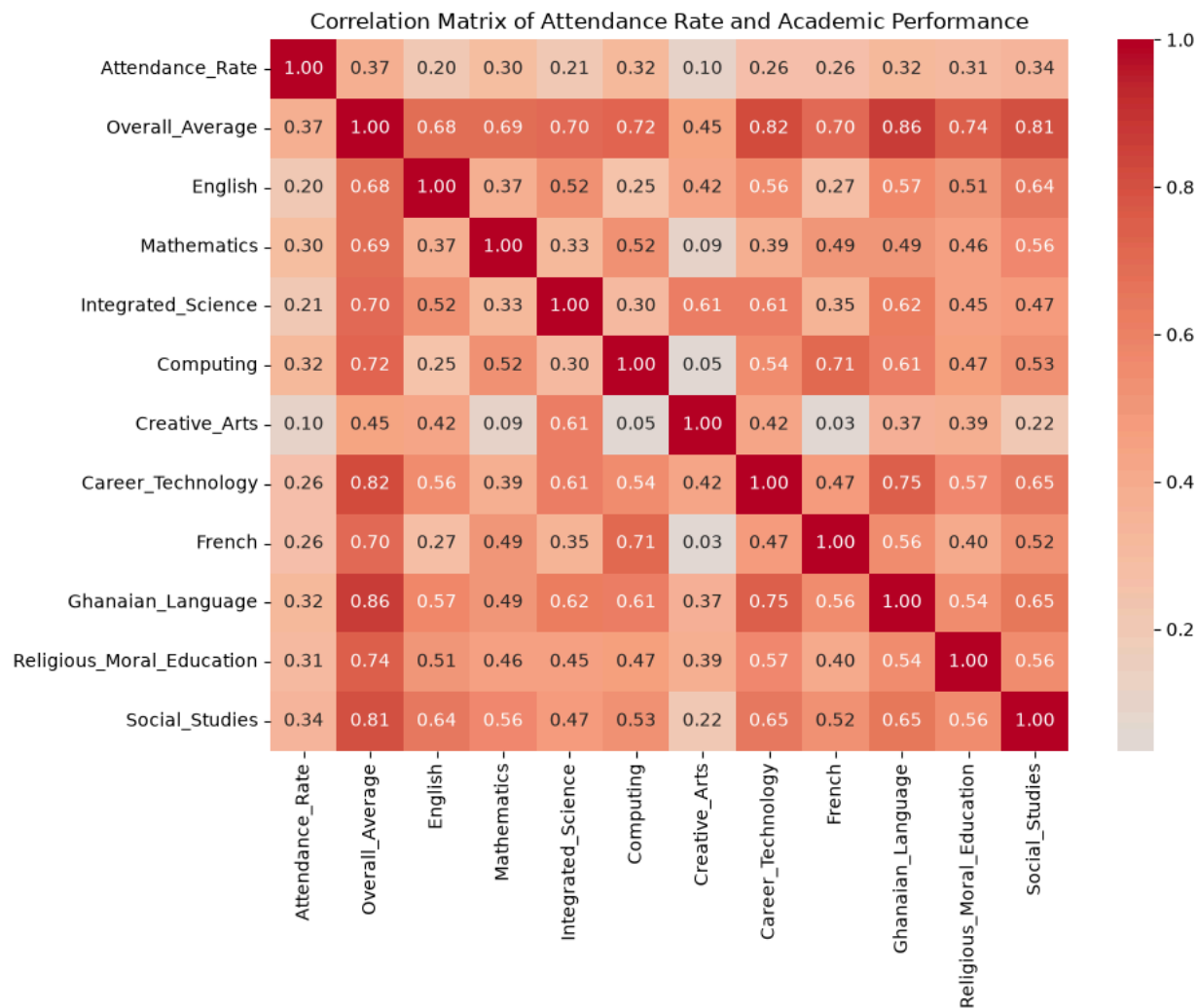
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## 5.1 Distribution of Student Performance by Subject



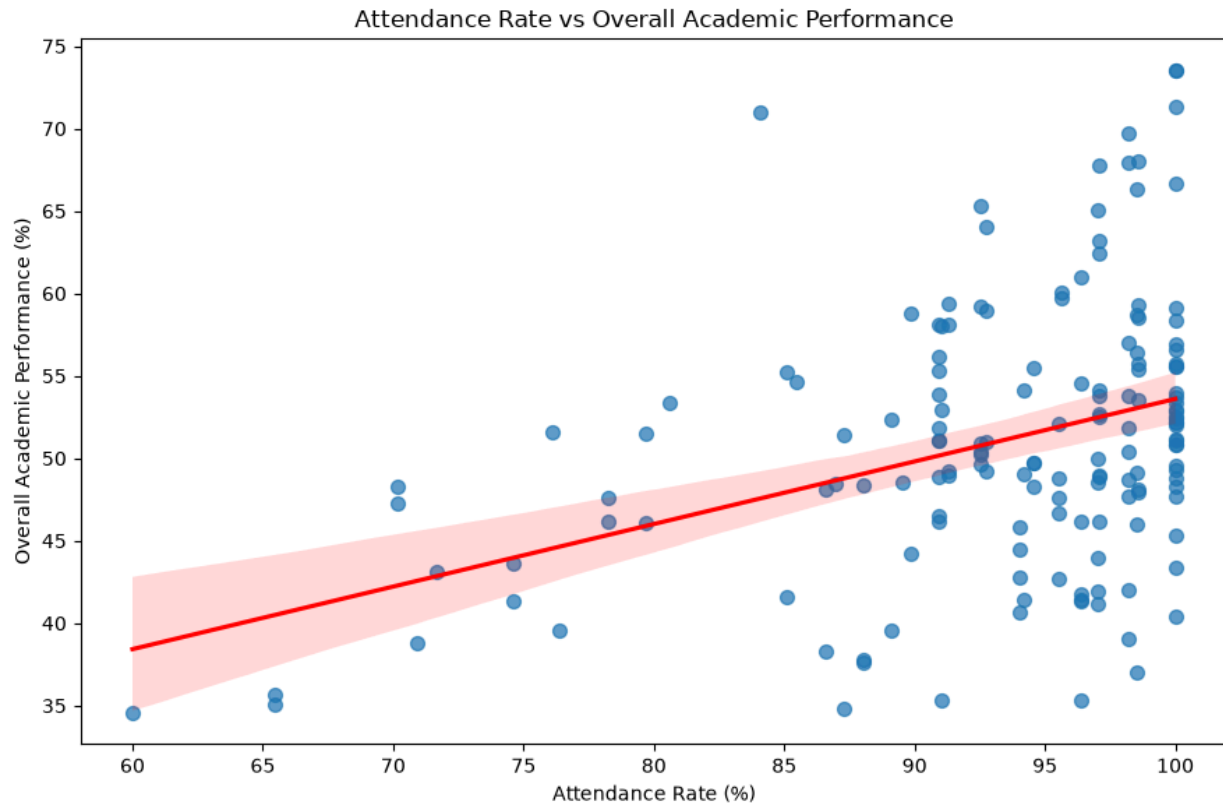
The box plot summarizes the distribution of student scores across all ten subjects. It reveals noticeable differences in both central tendency and score variability among subjects. Religious Moral Education and Social Studies exhibit the highest median scores, indicating relatively stronger overall student performance in these subjects. In contrast, Mathematics and Ghanaian Language record the lowest median scores, suggesting that students generally experienced greater difficulty in these areas. The spread of the boxes and whiskers also indicates varying levels of consistency across subjects. Mathematics and French display wider score distributions, reflecting greater variation in student achievement, while English and Creative Arts show comparatively more consistent performance. Several subjects contain high-performing and low-performing outliers, highlighting exceptional student performances as well as learners who may require additional academic support. Overall, this visualization provides an initial overview of student achievement patterns and identifies the subjects that warrant closer investigation in the subsequent analyses.

## 5.2 Correlation Between Attendance and Academic Performance



To better understand the relationships among attendance, overall academic achievement, and individual subject performance, a Pearson correlation analysis was conducted. Correlation coefficients range from -1 to +1, where values closer to +1 indicate a strong positive relationship, values near 0 indicate little or no relationship, and negative values indicate an inverse relationship. The correlation heatmap provides a comprehensive overview of how attendance and subject scores are associated with students' overall academic performance. By visualizing these relationships simultaneously, the analysis identifies which academic factors are most strongly linked to overall student achievement and assesses whether attendance alone is a significant predictor of academic success. The findings from this analysis guide the subsequent investigation into the relative influence of attendance and subject performance on overall student outcomes.

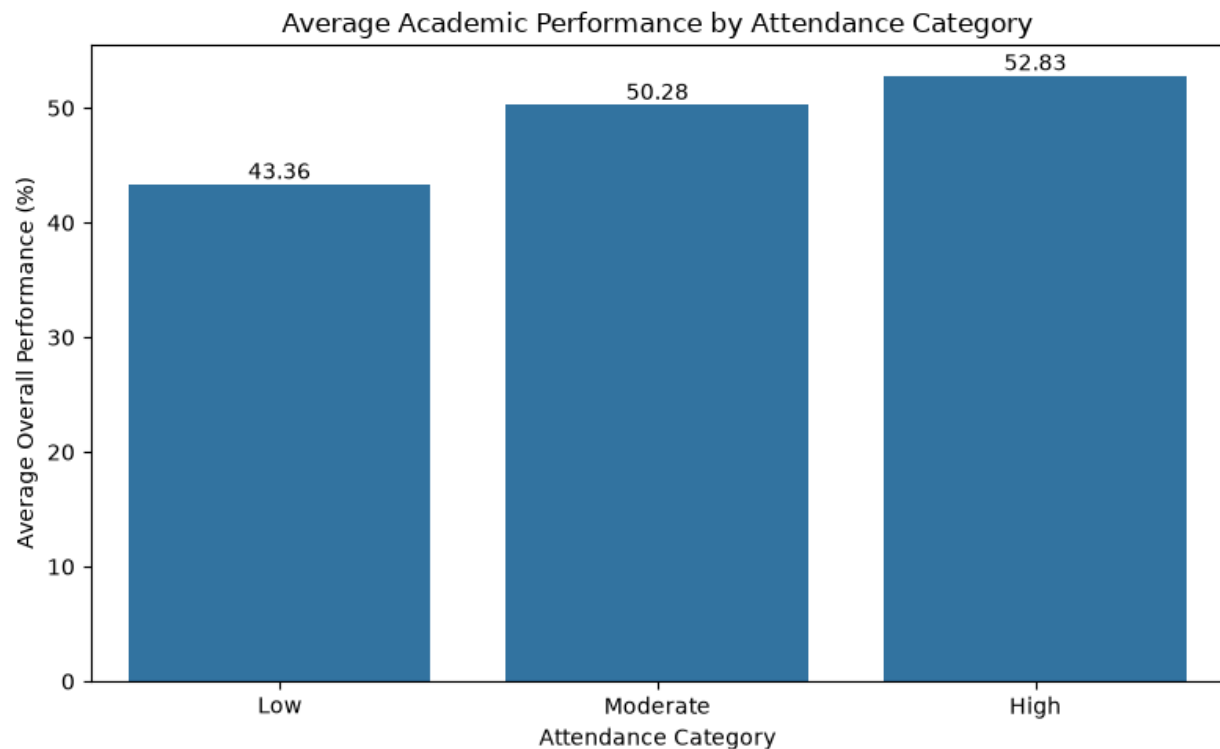
### 5.3 Relationship Between Attendance Rate and Overall Academic Performance



Following the correlation analysis, a scatter plot was generated to further examine the relationship between students' attendance rates and their overall academic performance. While the correlation matrix quantifies the strength of the relationship, the scatter plot provides a visual representation of how individual students' academic performance changes with varying attendance levels. The fitted regression line illustrates the overall trend within the data, while the distribution of individual data points reveals the degree of variability in student performance at different attendance rates. This visualization helps determine whether higher attendance consistently translates into improved academic outcomes or whether additional factors may also contribute to student achievement.



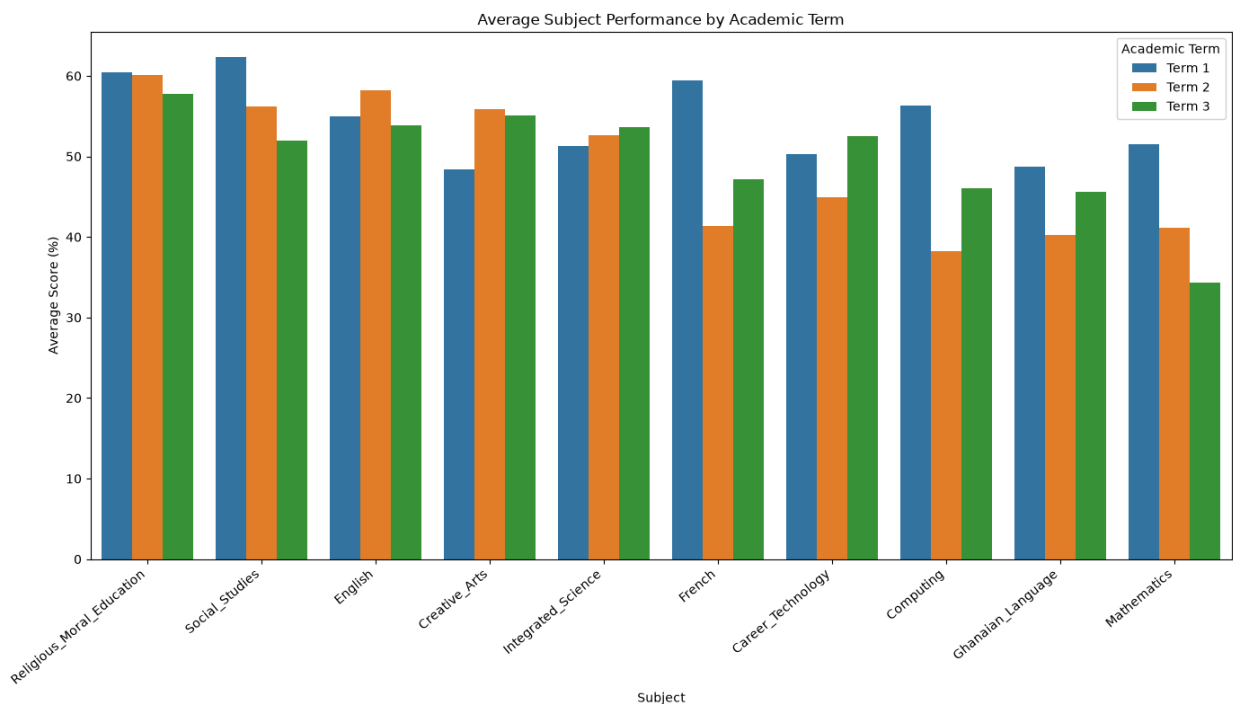
## 5.4 Average Academic Performance by Attendance Category



To further evaluate the impact of attendance on academic achievement, students were grouped into three attendance categories: Low, Moderate, and High, based on their attendance percentages. The average overall academic performance was then calculated for each category to determine whether improvements in attendance are associated with better academic outcomes. Grouping students into attendance categories provides a clearer comparison than analyzing individual attendance records alone. This approach highlights broad performance trends across different levels of attendance and helps assess whether consistently higher attendance is linked to improved academic achievement. The bar chart reveals a clear upward trend in academic performance as attendance levels increase. Students in the Low Attendance category achieved the lowest average overall academic performance (43.36%), while students in the Moderate Attendance category recorded a substantially higher average (50.28%). Students with High Attendance achieved the highest average performance (52.83%). The largest improvement occurs between the Low and Moderate attendance groups, indicating that maintaining a reasonable level of attendance is associated with meaningful gains in academic achievement. However, the smaller increase between the Moderate and High attendance categories suggests that once students consistently attend classes, further increases in attendance produce only modest improvements in performance. These findings complement the previous correlation and

scatter plot analyses. While regular attendance contributes positively to student achievement, it is not the search determinant of academic success. Other academic factors, including subject proficiency, assessment performance, and individual learning abilities, continue to play significant roles in influencing overall student outcomes.

## 5.5 Subject Performance Trends Across Academic Terms



To examine how student achievement evolved throughout the academic year, the average score for each subject was calculated and compared across the three academic terms. This analysis provides insight into whether students demonstrated consistent improvement, decline, or fluctuating performance in individual subjects over time. Comparing subject performance across terms enables educators to identify subjects with sustained achievement, subjects showing improvement, and subjects that may require targeted instructional support. These trends also help evaluate the effectiveness of teaching strategies and highlight areas where academic interventions may be necessary.

The grouped bar chart reveals noticeable variations in subject performance across the three academic terms, indicating that student achievement was not uniform throughout the academic year. Several subjects, including Creative Arts, Integrated Science, and Career Technology, showed gradual improvement from Term 1 to Term 3, suggesting that students made steady academic progress in these areas. In contrast, subjects such as Social Studies, French, Computing, and Mathematics experienced declines after strong performances in Term 1, with only partial recovery observed in some subjects during Term 3. Religious

Moral Education remained one of the strongest-performing subjects across all three terms, maintaining consistently high average scores. Conversely, Mathematics recorded the lowest average scores overall and exhibited the greatest decline by the final term, highlighting it as a subject that may require additional instructional support and targeted intervention. Overall, the analysis demonstrates that academic performance varies considerably across both subjects and terms. These findings emphasize the importance of monitoring subject-level trends throughout the academic year rather than relying solely on overall student averages, as different subjects exhibit distinct performance patterns that can inform curriculum planning and resource allocation.

## 6. Dashboard Development

A multi-page interactive dashboard was developed in Microsoft Power BI to communicate insights effectively. The dashboard included: The dashboard was connected to the Azure-hosted MySQL database through SQL views, enabling Power BI to retrieve analysis-ready datasets. This architecture reflects a modern cloud-based business intelligence workflow where database management, cloud infrastructure, analytics, and visualization are integrated into a unified reporting solution.

- Executive overview
- Academic performance analysis
- Attendance analysis
- Term progression
- Trends and insights

Interactive slicers enabled users to filter results by student and academic term.

## 7. SQL Analysis

MySQL served as the foundation of the analytical workflow and was deployed using Microsoft Azure Database for MySQL to provide a cloud-based data repository. The Azure deployment enabled centralized storage of educational data while supporting seamless connectivity with SQL clients, Python, and Power BI. The relational database transformed raw educational records into structured, analysis-ready datasets through normalized tables, SQL views, calculated fields, and analytical queries.

- **vw\_student\_performance** – Combined student records with subject scores and calculated each student's overall academic average and performance band.
- **vw\_attendance\_summary** – Normalized attendance by calculating attendance percentages based on the total instructional days in each academic term, allowing fair comparisons between terms of different lengths.
- **Subject Performance Views** – Aggregated subject-level performance to identify the highest and lowest-performing subjects.
- **Term Performance Views** – Summarized academic performance and attendance across the three academic terms to support trend analysis.

The SQL layer also generated several key performance indicators used throughout the dashboard, including:

- Overall academic average
- Attendance rate (%)
- Pass and fail rates
- Highest and lowest-performing students
- Subject rankings
- Performance bands
- Attendance categories

Creating reusable SQL views significantly reduced computational complexity in Power BI and Python by centralizing business logic within the database. This approach improved consistency, enhanced query performance, and established a scalable foundation for future educational reporting.

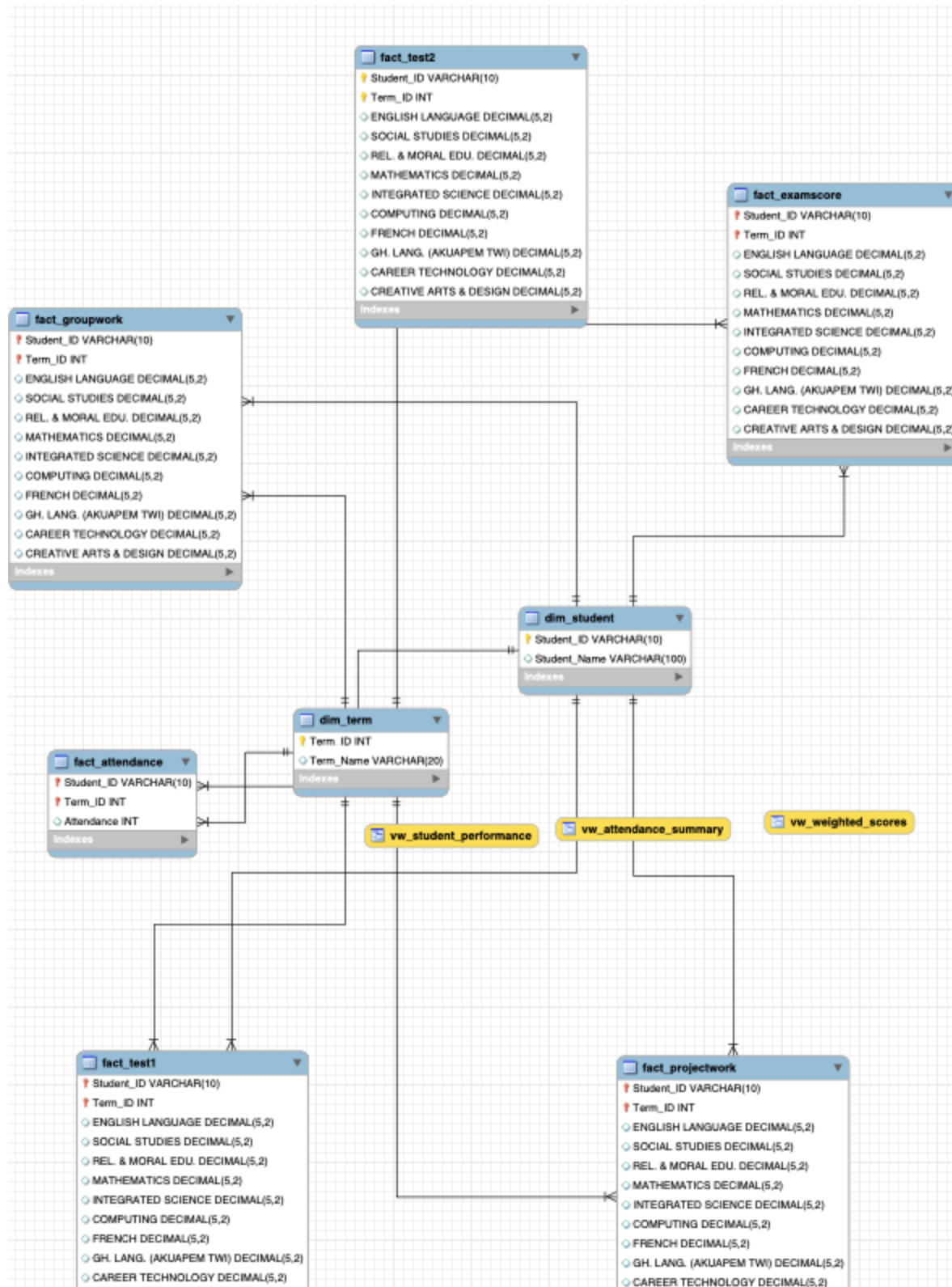


Figure: Entity-Relationship (EER) Diagram of the Educational Analytics database. The EER diagram illustrates the star-schema design comprising fact tables for assessments and

attendance, dimension tables for students and academic terms, and SQL views used as the primary analytical layer for Python and Power BI.

## 7. Exploratory Data Analysis

### 7.1 Overall Student Performance

The overall academic average across all students was approximately 51%. This suggests that while most students perform above the pass threshold, there is significant room for improvement.

### 7.2 Performance Across Terms

Analysis revealed a gradual decline in performance during Term 2 before recovering slightly in Term 3. This pattern suggests that additional investigation into mid-year learning challenges may be warranted.

- **Term 1:** 54.38% average score.
- **Term 2:** 48.88% average score.
- **Term 3:** 49.79% average score.

### 7.3 Subject Performance

There were clear disparities in achievement across different disciplines:

- **Highest Performance:** Religious and Moral Education (RME) recorded a mean of 59.46%.
- **Lowest Performance:** Mathematics recorded the lowest mean of 42.37% with a high standard deviation (14.25), indicating severe classroom polarization.

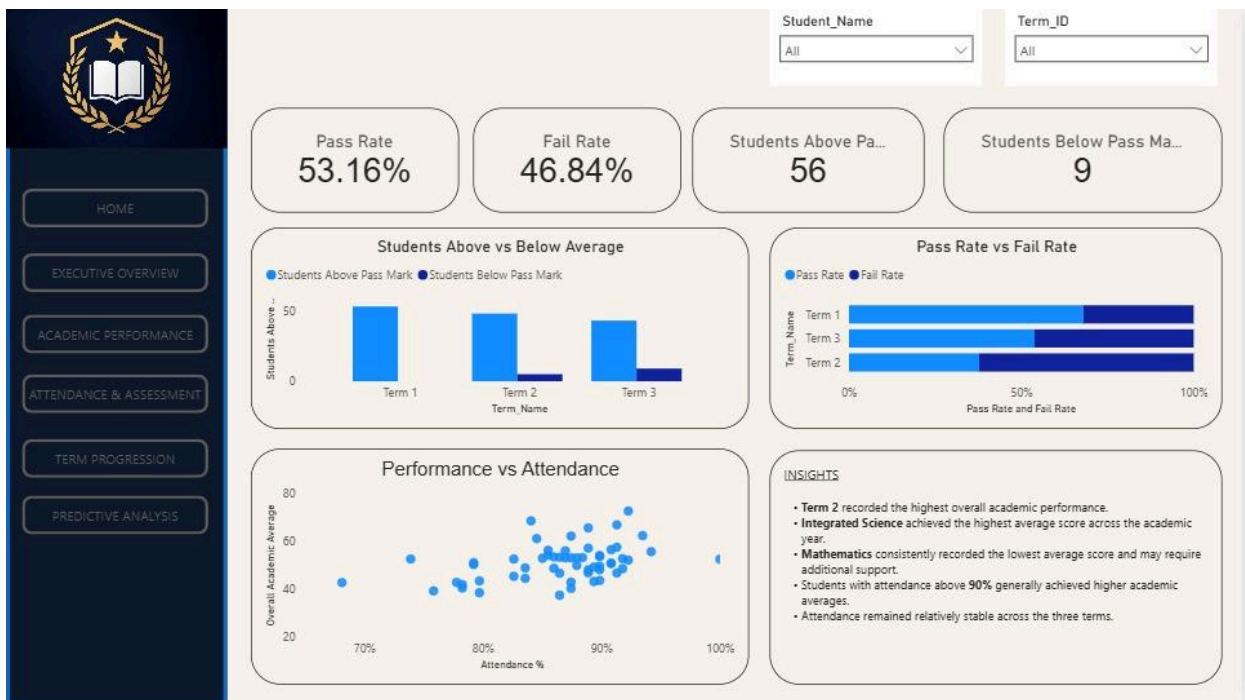
### 7.4 Attendance Analysis

Attendance remained consistently high, exceeding **92%** across all terms. Correlation analysis revealed a moderate positive relationship ( $r \approx 0.37$ ) between attendance and overall academic performance, indicating that students with higher attendance generally achieved better academic outcomes. However, attendance alone did not fully explain differences in performance.

## 8. Dashboard Development

The interactive Power BI dashboard provides stakeholders with a six-page narrative:

- **Home & Executive Overview:** High-level institutional indicators.
- **Academic Performance:** Subject-specific breakdowns and student rankings.
- **Attendance and Assessment:** Scatter plots mapping attendance against academic achievement.
- **Term Progression & Trends:** Longitudinal views of performance trajectories.



**Figure X. Executive Overview Dashboard**

The Executive Overview dashboard provides a high-level summary of institutional performance by consolidating key academic and attendance indicators into a single interactive interface. It presents the total number of students, the number of academic terms analyzed, the overall academic average, and the average attendance rate, allowing school administrators to quickly assess the institution's overall performance.

The dashboard also illustrates trends in academic performance and attendance across the three academic terms, highlights the distribution of students across performance bands, and identifies the highest-performing students based on their overall academic averages.

Interactive slicers enable users to filter the results by student and academic term, supporting dynamic exploration of the data and facilitating evidence-based educational decision-making.

## 9. Key Findings

### 9.1 Student Attendance Remained Consistently High

Student attendance remained consistently above **92%** across all three academic terms, indicating strong classroom participation and effective attendance management. However, high attendance alone did not guarantee strong academic performance.

### 9.2 Attendance Has Only a Moderate Influence on Academic Performance

Correlation analysis revealed a **moderate positive relationship** ( $r \approx 0.37$ ) between attendance and overall academic performance. Students with higher attendance generally achieved better academic results, but the relationship was not strong enough to conclude that attendance alone determines academic success. This suggests that additional academic, instructional, and learner-related factors contribute significantly to student achievement.

### 9.3 Overall Academic Performance Declined Across the Academic Year

Average academic performance was highest during **Term 1**, declined noticeably in **Term 2**, and recovered only slightly in **Term 3**. This trend indicates that student performance weakened as the academic year progressed and suggests that additional academic support may be beneficial during the middle of the school year.

### 9.4 Mathematics Emerged as the Weakest Subject

Mathematics consistently recorded the **lowest average score** among all subjects and exhibited one of the largest variations in student performance. The wide distribution of



scores suggests differences in student understanding and highlights Mathematics as a priority area for targeted instructional intervention.

## 9.5 Religious and Moral Education Demonstrated the Strongest Performance

Religious and Moral Education consistently achieved the highest average scores across all three academic terms. Together with Social Studies and English, it contributed positively to students' overall academic averages, suggesting effective instructional delivery and strong student engagement in these subjects.

## 9.6 Subject Performance Varied Considerably Across the Curriculum

The box plot and subject trend analyses revealed substantial differences in both average scores and score variability across subjects. While some subjects maintained relatively stable performance throughout the academic year, others experienced noticeable declines or improvements, emphasizing the importance of monitoring subject-specific trends rather than relying solely on overall academic averages.

## 9.7 Moderate Attendance Is Associated with Significant Performance Gains

Students in the **Moderate** and **High Attendance** categories achieved considerably higher average academic scores than those in the **Low Attendance** category. However, the improvement between Moderate and High attendance groups was relatively small, suggesting that once students maintain regular attendance, additional gains in performance depend more heavily on other academic factors.

## 9.8 Performance Trends Differ Across Subjects

Analysis of subject performance across academic terms revealed that several subjects, including **Creative Arts**, **Integrated Science**, and **Career Technology**, showed gradual improvement over time. Conversely, subjects such as **Mathematics**, **Computing**, and **French** experienced declines after strong performances in Term 1, indicating that subject-specific instructional strategies may require review.

## **9.9 Most Students Performed Within the Average Performance Range**

The distribution of overall academic averages showed that the majority of students clustered around the middle performance bands, with relatively few students achieving exceptionally high or exceptionally low overall averages. This suggests that school-wide improvement strategies may have a broader impact than interventions targeting only a small group of students.

## **9.10 The Dashboard Provides an Effective Decision-Support Tool**

The integration of SQL, Python, and Power BI resulted in an interactive dashboard capable of monitoring attendance, academic performance, subject trends, and student rankings in real time. The dashboard enables educators and administrators to identify emerging performance issues early and supports evidence-based educational planning.

# **10. Recommendations**

## **10.1 Strengthen Mathematics Instruction Through Targeted Intervention**

Mathematics consistently recorded the lowest average performance across all three academic terms while also exhibiting one of the highest levels of score variability among students. This indicates that many students struggle with foundational mathematical concepts and that learning outcomes are inconsistent across the cohort.

To address this challenge, the school should implement targeted intervention programs, including remedial classes, peer tutoring, differentiated instruction, and regular formative assessments. Teachers should also identify students who consistently perform below the class average and provide individualized academic support. Strengthening Mathematics instruction is expected to improve overall academic performance and reduce performance disparities among students.

## **10.2 Introduce Mid-Term Academic Support Programs**

Analysis of student performance revealed that average academic achievement declined noticeably after Term 1, with only a modest recovery observed during Term 3. This pattern suggests that many students begin to experience academic difficulties during the middle of the academic year.

Rather than waiting until end-of-term examinations, the school should introduce structured mid-term academic interventions, including progress monitoring, supplementary lessons, revision sessions, and targeted support for students identified as academically at risk. Early intervention will allow teachers to address learning gaps before they negatively affect end-of-term performance.

### **10.3 Promote Cross-Departmental Collaboration and Instructional Best Practices**

The analysis showed considerable variation in performance across subjects. Religious and Moral Education, Social Studies, and English consistently achieved stronger academic outcomes than Mathematics, Computing, and some technical subjects.

School leadership should encourage collaboration among departments by facilitating the sharing of successful teaching strategies, assessment methods, and classroom management practices. Professional learning communities and peer observations can help teachers adopt effective instructional approaches from high-performing departments, ultimately improving student outcomes across the curriculum.

### **10.4 Maintain High Attendance While Expanding Academic Support**

Although attendance remained consistently above 92% throughout the academic year, the analysis demonstrated only a moderate relationship between attendance and academic performance ( $r \approx 0.37$ ). This indicates that while regular attendance contributes positively to learning, it is not sufficient on its own to ensure academic success.

The school should continue implementing existing attendance initiatives while complementing them with academic support programs such as study skills workshops, mentoring, personalized learning plans, and regular academic feedback. These combined strategies are more likely to improve student achievement than attendance initiatives alone.

## **10.5 Implement Continuous Data-Driven Monitoring**

One of the key outcomes of this project is the development of an interactive Power BI dashboard capable of monitoring attendance, academic performance, subject trends, and student achievement in real time.

School administrators and teachers should integrate the dashboard into routine academic review meetings to monitor institutional performance, identify at-risk students, evaluate subject-level trends, and assess the effectiveness of intervention programs. Regular use of data-driven insights will support evidence-based decision-making and enable timely responses to emerging academic challenges.

## **10.6 Prioritize Subject-Specific Performance Reviews**

Subject trend analysis revealed that different subjects followed distinct performance trajectories across the academic year. While some subjects demonstrated gradual improvement, others experienced noticeable declines after Term 1.

School leadership should conduct periodic subject-level performance reviews to identify curriculum challenges, evaluate assessment practices, and allocate instructional resources where they are most needed. Monitoring performance at the subject level will enable more targeted interventions than relying solely on school-wide academic averages.

## **10.7 Establish an Early Warning System for At-Risk Students**

The dashboard and analytical models developed during this project provide an opportunity to identify students who consistently demonstrate low academic performance before they reach critical stages of the academic year.

The school should establish an early warning system that combines attendance records, assessment results, and overall academic averages to identify students requiring additional academic or pastoral support. Early identification will enable timely intervention, improve student retention, and reduce the likelihood of prolonged academic underperformance.

## **10.8 Foster a Sustainable Data-Driven Decision-Making Culture**

To maximize the long-term value of the analytics system, the school should encourage the routine use of educational data in planning, curriculum evaluation, and policy development.

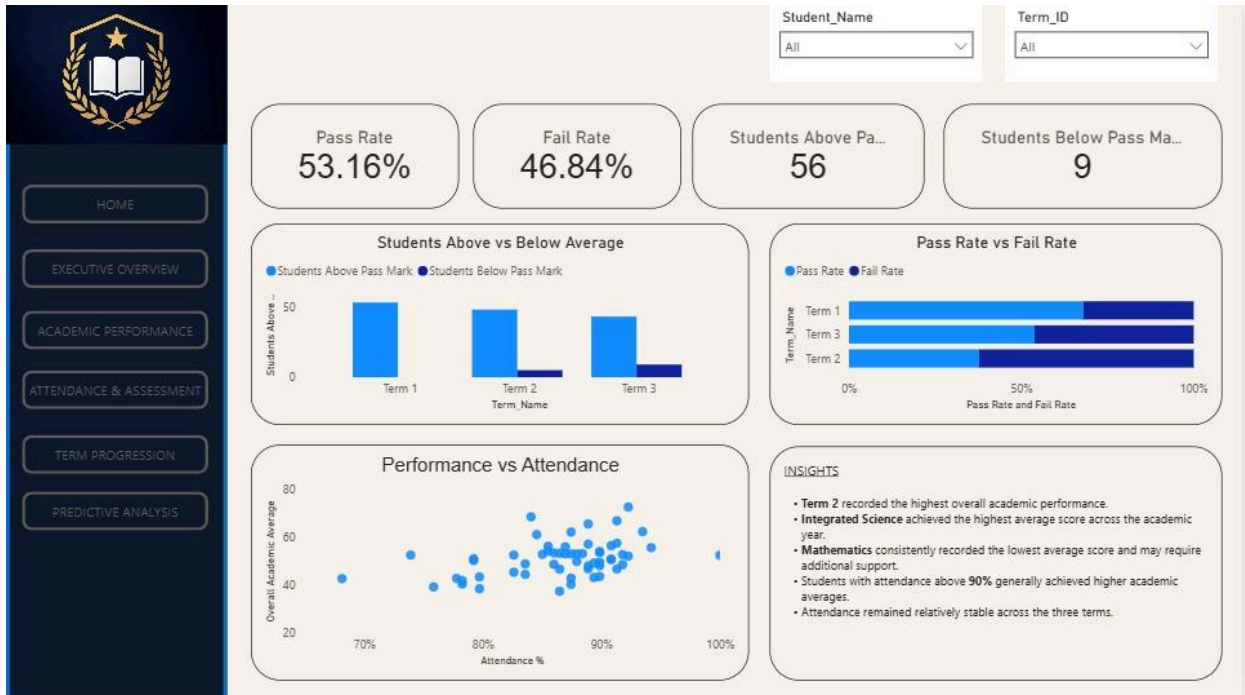
Administrators and teachers should receive training in interpreting dashboard metrics and using analytical findings to inform instructional decisions.

Embedding data-driven practices into the school's decision-making processes will support continuous improvement, enhance accountability, and contribute to better educational outcomes over time.

## **11. Conclusion**

This project illustrates a complete end-to-end educational analytics workflow. By integrating MySQL, Microsoft Azure, SQL, Python, and Power BI into a unified analytics pipeline, this project demonstrates how educational institutions can implement scalable, cloud-enabled business intelligence solutions that support evidence-based decision-making. The resulting platform transforms raw educational records into interactive dashboards that enable administrators to monitor performance, identify emerging risks, and improve student outcomes through data-driven interventions. The move toward data-driven decision-making ensures that educational resources are allocated where they are needed most, ultimately improving student outcomes across the Ghanaian educational ecosystem.

## **12. Appendix**





HOME

EXECUTIVE OVERVIEW

ACADEMIC PERFORMANCE

ATTENDANCE & ASSESSMENT

TERM PROGRESSION

TRENDS & INSIGHTS

Student\_Name

All

Term\_ID

All

Average Attendance

93.16

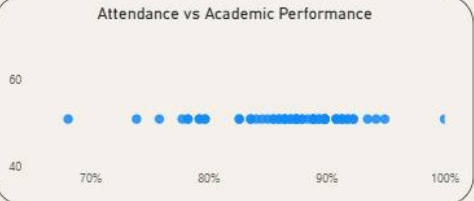
Highest Attendance

69

33

Lowest Attendance

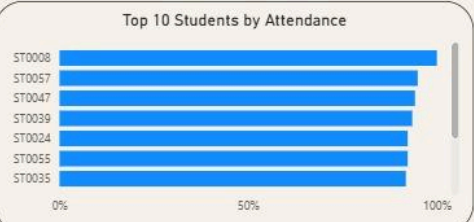
Attendance vs Academic Performance



Attendance by Term

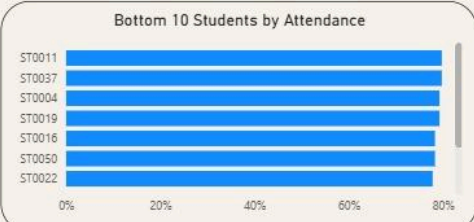


Top 10 Students by Attendance



| Student ID | Attendance (%) |
|------------|----------------|
| ST0008     | 95             |
| ST0057     | 94             |
| ST0047     | 93             |
| ST0039     | 92             |
| ST0024     | 91             |
| ST0055     | 90             |
| ST0035     | 89             |
| ST0011     | 88             |
| ST0037     | 87             |
| ST0004     | 86             |

Bottom 10 Students by Attendance



| Student ID | Attendance (%) |
|------------|----------------|
| ST0011     | 75             |
| ST0037     | 74             |
| ST0004     | 73             |
| ST0019     | 72             |
| ST0016     | 71             |
| ST0050     | 70             |
| ST0022     | 69             |
| ST0008     | 68             |
| ST0057     | 67             |
| ST0047     | 66             |



